

Project 5: Vehicle Fuel Consumption Prediction System

Automotive Powertrain Fuel Economy Regression System

1. Project Overview & Core Engineering Problem

Predicting vehicle fuel consumption (L/100 km & MPG) based on engine displacement, vehicle mass, horsepower, fuel type, and transmission specifications.

2. Mechanical & Thermodynamics Foundation

Based on vehicle inertial resistance ($E = m \cdot a$), engine displacement pumping losses, aerodynamic drag, fuel type offsets, and transmission friction.

3. Dataset Telemetry & Feature Specifications

Feature Name	Unit / Type	Engineering Significance
Engine Size	L	Engine displacement volume
Vehicle Weight	kg	Curb mass of the automobile
Horsepower	HP	Rated engine power output
Fuel Type	Categorical	Petrol, Diesel, Hybrid, Plug-in Hybrid
Transmission	Categorical	Automatic or Manual gearbox

4. Machine Learning Algorithms & Benchmark Performance

Model Architecture	Type / Task	Performance Benchmark	Engineering Evaluation
Random Forest Regressor	Regression	R²: 0.9714, MAE: 0.43 L	Top accuracy fuel economy predictor
Linear Regression	Regression	R²: 0.9622, MAE: 0.50 L	Strong baseline parametric model
Decision Tree Regressor	Regression	R²: 0.9466, MAE: 0.60 L	Rule-based tree regressor

5. Student Curriculum & Educational Takeaways

- Mastering One-Hot Encoding and StandardScaler preprocessing with ColumnTransformer.
- Converting metric fuel consumption (L/100 km) to imperial fuel economy (MPG).
- Calculating annual fuel expenditure (\$/year) and CO2 carbon footprint.